

# Anh Thuan Tran

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## SUMMARY

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PhD student @ GMU | 3D Vision, RGB-D SLAM, 3D Gaussian Splatting, Neural Scene Representations.

## EDUCATION

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### George Mason University

Ph.D in Computer Science | Advisor: Prof. Jana Kosecka

Virginia, USA

Sep 2024 – Exp. May 2029

### Pukyong National University

M.Eng. in Artificial Intelligence Convergence | GPA: 3.89

Busan, South Korea

Mar 2022 – Feb 2024

### Vietnam National University

B.Ec. in E-commerce | GPA: 3.42 (Thesis track)

Ho Chi Minh City, Vietnam

Sep 2017 – Jul 2021

## PUBLICATIONS

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- [1] **Anh Thuan Tran** and Jana Kosecka. VarSplat: Uncertainty-aware 3D Gaussian Splatting for Robust RGB-D SLAM. In *CVPR*, 2026. [\[PDF\]](#) | [Project Page](#).
- [2] **Anh Thuan Tran** and Jana Kosecka. PointSplat: Efficient Geometry-Driven Pruning and Transformer Refinement for 3D Gaussian Splatting. In *Under Review (CVPRW)*, 2026. [\[PDF\]](#).
- [3] **Anh-Thuan Tran**, Hoanh-Su Le, Suk-Hwan Lee, and Ki-Ryong Kwon. PointCT: Point Central Transformer Network for Weakly-Supervised Point Cloud Semantic Segmentation. In *WACV*, pages 3556–3565, 2024. [\[PDF\]](#) | [Code](#).

## RESEARCH EXPERIENCE

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### Artificial Intelligence and Database Lab

Graduate Research Assistant

GMU, USA

Sep 2024 – Present

- **VarSplat** (SLAM & Uncertainty): Built an uncertainty-aware RGB-D 3DGS-SLAM system, rendering per-pixel uncertainty for robust tracking and mapping under challenging conditions.
- **PointSplat** (Pruning): Developed geometry-driven pruning and refinement for 3D Gaussians in medium-scale indoor scenes with Point Transformer network.
- **Dynamic 3DGS** (Ongoing): Developing uncertainty-aware 3DGS for both short-term and long-term changes in online and monocular settings.
- **Sparse Reconstruction** (Ongoing): Building method that constrains generative priors with underlying geometry through uncertainty modeling for 3DGS sparse-view reconstruction.

### Multimedia Signal Processing Lab

Graduate Research Assistant

PKNU, South Korea

Mar 2022 – Feb 2024

- **PointCT**: Proposed point-based Transformer network for weakly-supervised point cloud segmentation.
- Developed central-based attention block to integrate global features across neighborhoods, and improve unlabeled point representations.

## SKILLS

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- **Programming:** Python
- **Frameworks:** PyTorch, OpenCV, NumPy
- **3D Vision/Graphics:** 3D Gaussian Splatting, NeRF, RGB-D SLAM, Point Clouds, Point Transformers, Differentiable Rasterization, NeRFStudio
- **Systems:** Linux/Ubuntu, Slurm, Multi-GPU training
- **Vision Foundation Models:** CLIP, DINOv2, SAM
- **Other:** PostgreSQL, FAISS

## SCHOLARSHIPS

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- Graduate Teaching Assistant, George Mason University (Sep 2024 - Present)
- Brain Korea 21, Graduate Student Research Scholarship (Artificial Intelligence Convergence Research Group), Pukyong National University (Mar 2022 - Feb 2024)
- Full-time Scholarship For Graduate Student, Pukyong National University (Mar 2022 - Feb 2024)
- Graduate student scholarship, Pukyong National University (Sep 2022 - Feb 2023)

## WORK EXPERIENCE

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|----------------------------------------|-------------------------------------------|
| <b>Newgen I&amp;S</b><br>AI Researcher | Busan, South Korea<br>Jan 2024 – Jul 2024 |
|----------------------------------------|-------------------------------------------|
- Developed offline chatbots for marine industry using LangChain and Ollama to process bilingual (English–Korean) PDFs.
  - Processed text with FAISS, updated PostgreSQL, and LLM responses with top-k retrieval.
  - Deployed system on server with WebSocket for real-time communication.

## AWARDS

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- Outstanding International Student Award - Pukyong National University (2024)
- Excellent achievement in undergraduate research – VNU-HCM (2022)
- Third Prize Young Scientist – University of Economics and Law, VNU-HCM (2018, 2020)